

Preet Raval

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PROFESSIONAL SUMMARY

DevOps and full-stack engineer who has been the sole engineer behind five production systems used daily by 50+ employees at American Circuits Inc., covering traveler management, inventory, internal AI tooling, and the infrastructure behind all of it. I own systems end-to-end, from schema design and REST APIs through UI, deployment, and uptime. Core stack is Python, FastAPI, Next.js, PostgreSQL, Docker, and Linux, with hands-on experience in CI/CD automation, LLM and OCR integration, and enterprise SSO/MFA security.

SKILLS

Languages & Frameworks: Python, TypeScript/JavaScript, Java, C#/.NET, React, Next.js, FastAPI, Java Spring Boot, Tailwind CSS, HTML5/CSS3, RESTful APIs, Microservices

Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Oracle Database, BigQuery, SQL Optimization, Data Modeling, ORM

Cloud & DevOps: AWS, Azure, Docker, Kubernetes (Amazon EKS), Terraform, ArgoCD, GitHub Actions, Nginx, Vercel, Cloudflare Tunnels, Linux/On-Premise Infrastructure, Server Administration

Observability: Prometheus, OpenTelemetry, Elasticsearch

AI/ML: LLM Integration (Claude API, ChatGPT API), Prompt Engineering, RAG, OCR, Agentic AI, Model Context Protocol (MCP), TensorFlow, PyTorch, Hugging Face Transformers, Pandas, NumPy

Security & Networking: SSO/MFA, Role-Based Access Control, MITRE ATT&CK, Penetration Testing, Vulnerability Assessment, SIEM, TCP/IP, Subnetting, VLANs, Encryption, IAM

Other: ERP/MRP/CRM (Odoo), Jira, Git, Agile/SDLC, Data Analytics, Tableau, Technical Documentation

EXPERIENCE

DevOps Engineer – AI Systems

Mar 2026 – Present

American Circuits Inc.
NC

Charlotte,

- Scaled and now run the Linux-based production infrastructure behind 5 business-critical applications used daily by 50+ employees across multiple environments.
- Built out CI/CD pipelines in GitHub Actions that cut manual deployment steps and shortened release turnaround time across the board.
- Containerized and orchestrated our microservices with Docker and Kubernetes across all 5 production systems, getting us to zero-downtime deployments.
- Hardened our security posture end-to-end – RBAC, secrets management, and infrastructure hardening aligned with best practices for business-critical systems.
- Automated reporting, operational workflows, and third-party integrations with Python and REST APIs, cutting hours of manual work every week.
- Own backups, monitoring, patching, and incident response, keeping production systems at 99%+ uptime.

Software Engineer

Jun 2025 – Mar 2026

American Circuits Inc.
NC

Charlotte,

- Built and shipped enterprise full-stack applications end-to-end, from design through production deployment, using Next.js, FastAPI, PostgreSQL, Docker, and Nginx.
- Designed backend APIs, database schemas, and deployment architecture used by engineering, manufacturing, and quality control teams company-wide.
- Set up and owned authentication infrastructure, including SSO, MFA, and zero-trust network access over Cloudflare Tunnel.
- Tracked down slow database queries and API bottlenecks and cut average response times by roughly 40% across our highest-traffic services.
- Partnered with QA on cross-functional bug fixes: traced issues to root cause, wrote regression tests, and shipped fixes without disrupting end users.
- Delivered a real-time ops dashboard for supervisors covering job status, queue depths, operator activity, and QC sign-off, and rolled out QR/barcode scanning to enforce step-by-step routing on the production floor.
- Deployed and maintained 4 on-premises servers (database, staging/UAT, and two application servers) with

health checks and uptime monitoring supporting concurrent production shifts.

Frontend Developer Intern

Jan 2022 – Dec 2022

Skillship Foundation

Remote

- Built responsive UI components across several product modules in React and Next.js, keeping the codebase modular and reusable.
- Independently translated Figma mockups into pixel-accurate, cross-browser layouts with Tailwind CSS, with consistent rendering across desktop, tablet, and mobile.
- Built reusable form components with client-side validation for registration, login, and course enrollment, which cut down incomplete submissions.
- Replaced hand-updated static pages with live REST API integration for course listings, user progress, and event schedules.
- Tracked down and fixed cross-browser rendering bugs in Safari and Firefox, tracing them to vendor-prefixed CSS instead of patching over them with brittle overrides.

PROJECTS

Pashupatastra – AI-Native Autonomous Security Operations Platform

Software Engineer

Stack: Next.js, TypeScript, Python, FastAPI, Kubernetes (EKS), Docker, Terraform, AWS, Prometheus, OpenTelemetry, Elasticsearch

- Built a closed-loop security operations platform that correlates fragmented alerts across tools into single incidents with causal chains and MITRE ATT&CK mappings, then proposes risk-scored, human-authorized remediation.
- Designed an Observe → Understand → Predict → Decide → Act → Verify → Learn workflow for autonomous incident response.
- Weighted response actions by blast radius, confidence, and reversibility, with dry-run mode, policy engine gates, and documented rollback plans guarding every automated action.
- Built a real-time Observatory that tracks ransomware disclosures, active botnet infrastructure, and CISA KEV exploited vulnerabilities on an hourly cadence.
- Architected the system as six modules: Drishti (perception), Smriti (memory), Buddhi (reasoning), Astra (remediation), Kavach (detection), and Dharma (policy).

Vyne – Unified Business & Infrastructure Event Timeline

Software Engineer

- Built an AI-native correlation layer that unifies business and infrastructure events from every connected tool into a single timeline.
- Owned the core backend API in FastAPI and PostgreSQL, handling real-time event ingestion from multiple third-party platforms at scale.
- Designed RBAC and tenant-level data isolation across a shared multi-tenant infrastructure to keep customer data properly bounded.
- Integrated GitHub, Sentry, Datadog, HubSpot, Stripe, Linear, and PagerDuty into one unified event timeline.
- Shipped 15+ product modules, including chat, project management, docs, voice/video calls with live transcription, and AI-powered meeting recaps.

Serenity Living – Client Website

Software

Engineer

- Took full ownership from brief to launch: UI/UX design, backend services, and Vercel deployment.
- Optimized Core Web Vitals through image compression, script deferral, and per-route code splitting, hitting a 95+ Lighthouse score on mobile.
- Built server-side logic for contact forms, newsletter signups, and booking requests through Next.js API routes, keeping sensitive processing off the client.
- Added Open Graph tags, JSON-LD structured data, and an auto-generated sitemap to boost search and social visibility.
- Built the CMS-driven content structure myself so the client can add pages, swap images, and update copy without needing an engineer – zero engineering incidents since launch.

EDUCATION

UNC Charlotte – Bachelor's, Computer Science

Jan 2023 – Dec 2025

Charlotte, NC

Vellore Institute of Technology – B.Tech, Computer Science (Transfer)

Sep 2020 – Dec 2022

India

CERTIFICATIONS

- AWS Solutions Architect (Associate), Amazon Web Services – In Progress
- Software Architecture (5-Level Track), O'Reilly by Pearson – In Progress
- Google Data Analytics Professional Certificate, Coursera/Google – May 2025
- AWS Cloud Solutions Architect, Coursera/AWS – Jun 2025
- Networking Fundamentals and Physical Networks, Coursera – Jun 2026
- Claude Code In Action, Anthropic – Apr 2026
- Claude with the Anthropic API, Anthropic – Aug 2026
- AI Fluency: Framework & Foundations, Anthropic – Aug 2026
- Introduction to Model Context Protocol, Anthropic – Aug 2026
- Introduction to Penetration Testing, O'Reilly by Pearson